

Brain drain

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In the new information economy that emerged in the 1990s, “human capital” devoted to creativity and innovation became the most critical resource that can ensure a country’s dominant presence in the world. This is reflected in the selective immigration policies of countries in the Organization for Economic Co-Operation and Development (OECD) and of developing and transition economies that favor the migration of skilled labor. These migration policies – including the US employment (H1B) visa program of the US government – increased the migration of highly skilled and educated individuals from developing nations to nations with employment opportunities in the information technology (IT) and science and technology fields, especially during 1990s. This process is referred to as “brain drain” because the sending countries in many instances lose a considerable number of their highly skilled people. Lowell (2003) states that two conditions are necessary for the phrase “brain drain” to apply to a given country: there must be a significant loss of the highly educated population; and adverse economic consequences must follow.

During the 1950s and 1960s the British Royal Society began using the expression “brain drain” to refer to the departure of scientists and technologists to the United States and Canada. The concept of brain drain also includes well-qualified technical experts, administrators, senior managers, doctors, engineers, investors, educators, businessmen, some workers in key positions, and subcontractor workers (Gençler & Çolak 2002). After leaving their countries to study abroad, university graduates and highly skilled individuals often do not return to their homeland but prefer to live and work in that country (Tansel & Gungor 2002).

Worldwide estimates of the numbers of immigrants contributing to the brain drain are

not available, but reliable estimates of immigrants and of their educational status in OECD countries show these countries to be significant destinations for highly skilled migrants. The United States has emerged as the world’s single largest magnet for foreign skilled workers: 40 percent of its foreign born adult population has tertiary level education. Since the early 1990s, approximately 900 000 highly skilled professionals, mainly IT workers, have migrated to the US, taking advantage of the temporary H1B visa program. These professionals are from India, China, Russia, and a few OECD countries – including Germany, the United Kingdom, and Canada. The United States admits 32 percent of all the foreign students studying in OECD countries. Higher education has proven to be an important channel for American firms to recruit highly skilled immigrants. Around 25 percent of H1B visa holders in 1999 were students previously enrolled in US universities. However, Canada also attracts talent and, despite its modest loss of skilled migrants to the United States, it is itself a net importer of human capital. Migration of skilled workers to France and Germany increased following the implementation of policies to attract researchers, IT workers, and foreign students. In 2000, Germany introduced a program similar to the American “green card” to recruit 20 000 foreign IT specialists. By the end of that year Germany had recruited half of that number, mostly from Eastern European countries (Cervantes & Guellec 2002). At present the scale of the brain drain is massive and global in scope. Demographer B. Lindsay Lowell and geographers Allan Findlay and Emma Stewart (2004; cited in Sriskandarajah 2005) say that nearly one in ten adults with some university or post-secondary schooling born in the developing world now lives in the developed world.

The phenomenon of the brain drain has been traditionally seen as having negative consequences for the less developed (LDC)

sending countries, which bear the cost of raising and educating professionals in their unproductive years. The departure of highly educated persons results in the loss of human capital as well as in the shrinkage and elimination of externalities, which the employment of the highly educated creates in the home country (Glytos 2010; Tessema 2010). These externalities include an understaffed health and education sector and, more significantly, a decline in their influence on social policies and institutions (Ozden & Schiff 2006). According to the African Capacity Foundation, African countries lose 20 000 skilled personnel to the developed world annually, which brings the number of immigrants in Europe and North Africa to over 300 000. Of these, about 30 000 are PhD holders (Tessema 2010). The developed world's efforts to increase aid for these countries may be futile if the local personnel required to implement development programs are absent. Small developing countries are especially vulnerable. Recent data from OECDs show that, in countries such as Guyana, Jamaica, Haiti, Trinidad and Tobago, and Fiji, two-thirds of all highly skilled workers have left these countries. The sheer volume of emigration from these small countries indicates that any possible positive consequence of aid may be surpassed by negative impacts on economic dynamism, the execution and delivery of key public services, and the depletion of the political classes. On the other hand, in larger countries like China and India, the relative scale of loss of human capital is smaller; however, particular sectors might be affected adversely, as for instance through the departure of huge numbers of health care workers from some sub-Saharan African countries (Lowell 2003; Sriskandarajah 2005).

The view that the poor sending countries bear the cost of education and the richer host countries enjoy the benefits of the generated human capital is not entirely true. Indeed this traditional view of brain drain is now being contested by scholars and is thought to be outdated and simplistic in the globalized economy. The brain drain premise has given way to the theory of brain gain/brain circulation, which

results from the presence of educated migrants in the host country: such a presence is a catalyst for the promotion of trade, capital flow, and technology transfers to their home country. This is especially true in the case of Asian countries like China, Taiwan, Korea, Singapore, and India, which have experienced rising economic growth since 1980s and 1990s. As these countries make technological progress and are changing towards becoming knowledge-based economies, they have embraced policies similar to those of the developed countries in order to compete for talent – including from the pool of their own expatriates. For these sending countries of the developing world, capturing the benefits mostly depends on attracting the skilled emigrants back to their homelands by offering them opportunities to use their new technological competencies. Returnees also bring valuable management experience, venture capital, entrepreneurial skills, and access to global networks. The relative success of China, Taiwan, Korea, and India in encouraging return migration has been attributed to the opening of their economies and to neoliberal state policies that foster domestic investments in innovation and research and development (R&D) (Saxenian 2005; Ray 2010). For instance, one of the Taiwanese government's most celebrated accomplishments has been the establishment of Hsinchu Science-based Industrial Park in 1980, where the government provided financial incentives and planned infrastructure for foreign companies to relocate or to form in this area. By 1987, 20 percent of the executives of large Taiwanese firms were former migrants (O'Neil 2003). Similarly, to encourage Indian professionals in the United States, who have been primary drivers of knowledge and capital flow to India, the Indian government has supported the emergence of private networks through legislative and tax breaks that encourage remittances and investments from Indians abroad (Cervantes & Guellec 2002). Mahroum writes: "the motion of brain drain and brain gain might be going in all directions: north–north, north–south, south–north, and south–south" (2000: 23).

Therefore the new brain drain literature deals with the brain drain-induced “brain gain” perspective and makes room for understanding the connection between sending and receiving countries not as a relationship of losers and winners, but as a global circulation of population and labor that contributes to changing the profile, perceptions, and actions of contemporary migrants. The idea of tapping the diaspora for resources has been put to use by advanced countries as well – countries like Switzerland, whose online network Swiss-List.com was established to promote networking among Swiss scientists in the United States and to encourage contacts with peers in Switzerland (Cervantes & Guellec 2002).

Further, Lowell, and Findlay (2002), Stark (2002), and others contend that “optimal brain drain” theory supports the notion that the prospects of being able to emigrate may increase the incentives to obtain education and skills and encourage additional investment in education. When the domestic “brain gain” is greater than “brain drain,” the net impact on welfare and growth may be rewarding, particularly in the case of poor countries. With “brain drain,” the average educational level of those who remain in the country may be higher than it would be without migration. Even though this theory is not proven beyond doubt, it is important to note that “brain drain” need not always have a negative impact on the sending country. The International Monetary Fund (IMF) estimates that, in 2004, feedback from expatriates also included US\$ 126 billion in remittances to less developed countries (OECD 2005). It exceeded the official development aid or incomes from the export of goods and services in several of these poor countries. This issue of remittances and economic growth has drawn the attention of several international organizations (IMF, OECD, World Bank) at a time when the volume of international aid is shrinking.

Brain drain has significant effects, whether they are positive or negative. The effects are context specific and calculating and managing the overall impacts of migration and brain drain is complicated to say the least. It calls

for methodological innovations to establish effective policy interventions that can create a “brain bank” from “brain drain.” The research challenges include devising methodologies for differentiating “brain drain” and “brain strain.” Sriskandarajah (2005) defines brain strain “as cases in which the net outflow of highly skilled workers from a particular sector in a particular country is actually hampering or is very likely to hamper economic development or the pursuit of important socio-economic gains.” Thus, being able to single out the “brain strain” hot spots, especially those that sabotage the efforts of achieving the United Nation’s Millennium Development Goals or the efforts to eliminate poverty in the world’s poorest countries, will be crucial for policy-makers, both in sending and in receiving countries, who are working to implement immigration policies that optimize the flow of skilled migrants.

SEE ALSO: Dependency theory; Developed and less developed societies; Higher education; Inequality, education; Migration; Networks; Remittances and development.

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